Conducting Material

- Conductors are the material that allows passage to heat, electricity, or sound through them.
- The material's ability to conduct electric current is called electrical conductivity.
- Electrical conductivity gives the value to which extent the material can conduct.
- The reciprocal of the conductivity is resistivity.

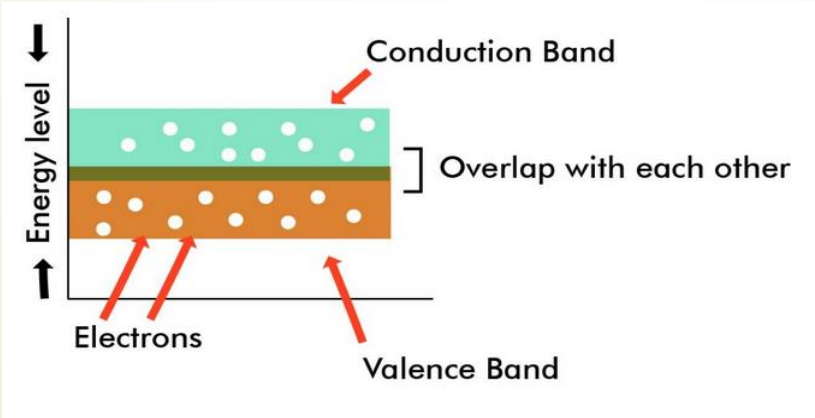
Conducting Material

How an electrical conductor conducts electricity?

- Electricity is the flow of charges. In a conductor, there are millions of atoms that possess millions of electrons.
- The electrons present in the outermost shell are called free electrons.
- These free electrons move randomly inside the conductor.
- But when an external voltage is applied, a potential difference is formed between the negative and positive terminal of the voltage source.
- Then the negatively charged electrons start moving from the negative terminal to the positive terminal.
- This starts conducting electric current in the direction opposite to the flow of electrons.

Energy band of the conductor

- Conduction of current is possible because of charge carriers (electrons).
- In the conductor, the charge carriers are present in the valence band.
- The valence band and the conduction band of the conductor overlap each other, therefore there is no forbidden energy gap between CB and VB



Energy band of the conductor

- So, a small amount of external energy provides enough energy to the electrons present in the valence band to move to the conduction band easily.
- At room temperature the conduction band of the conductor is almost full that's why the conductor easily conducts electricity at room temp.

Factors effecting the resistivity of electrical materials

1. Temperature.
2. Alloying.
3. Mechanical stressing.
4. Age Hardening.
5. Cold Working.

1. Temperature

- ✓ The resistivity of materials changes with temperature.
- ✓ Resistivity of most of the metals increase with temperature.
- ✓ The change in the resistivity of material with change in temperature is given by: $\rho_{t_2}=\rho_{t_1}[1+a_1(t_2-t_1)]$

Factors effecting the resistivity of electrical materials

- The resistivity of metals increase with increase of temperature.
- Means the metals are having positive temperature coefficient of resistance.
- Several metals exhibit the zero resistivity at temperature near to absolute zero.
- This phenomenon is “called the superconductivity”.
- The resistivity of semiconductors and insulators decrease with increase in temperature.
- Means the semiconductors and insulators are having negative temperature coefficient of resistance.

Factors effecting the resistivity of electrical materials

2. Alloying

- Alloying is a solid solution of two or more metals.
- Alloying of metals is used to achieve some mechanical and electrical properties.
- The atomic structure of a solid solution is irregular as compared to pure metals.
- Due to which the electrical resistivity of the solid solution increases more rapidly with increase of alloy content.
- A small content of impurity may increase the resistivity metal considerably.
- Even the impurity of low resistivity increases the resistivity of base metal considerably.
- For example the impurity of silver (having lowest resistivity among all metals) in copper increase the resistivity of copper.

Factors effecting the resistivity of electrical materials

3. Mechanical Stressing

- Mechanical stressing of the crystal structure of material develops the localized strains in the material crystal structure.
- These localized stains disturb the movement of free electrons through the material.
- Which results in an increase in resistivity of the material.
- Subsequently, annealing, of metal reduces the resistivity of metal.
- Annealing of metal, relieve the mechanical stressing of material due to which the localized stains got removed from the crystal structure of the metal.
- Due to which the resistivity of metal decrease.
- For example, the resistivity of hard drawn copper is more as compared to annealed copper.

Factors effecting the resistivity of electrical materials

4. Age Hardening

- Age hardening is a heat treatment process used to increase the yield strength and to develop the ability in alloys to resist the permanent deformation by external forces.
- Age hardening is also called “Precipitation Hardening”.
- This process increases the strength of alloys by creating solid impurities or precipitate.
- These created solid impurities or precipitate, disturb the crystal structure of metal which interrupts the flow of free electrons through metal/Due to which the resistivity of metal increases.

Factors effecting the resistivity of electrical materials

5. Cold Working

- Cold working is a manufacturing process used to increase the strength of metals.
- Cold working is also known as “Work hardening” or “Strain hardening”.
- Cold working is used to increase the mechanical strength of the metal.
- Cold working disturbs the crystal structure of metals which interfere with the movement of electrons in metal, due to which the resistivity of metal increases.

Classification of electrically conducting materials

1. Low resistivity materials
2. High resistivity materials

1. Low resistivity conducting materials

- Low resistivity materials or high conductivity materials mean those materials which have high conductivity or low resistivity.
- Generally good electrical conductors are this type of material.
- Low resistivity materials are used in transformer winding, in-house wiring as conductors for power transmission and distribution, and in machines like motors, generators, etc.
- Copper, aluminium, gold, and silver are examples of low resistivity materials.
- But commercially copper and aluminium are most commonly used.
- Silver and gold both are also good conductors of electricity but due to their high prices, they are not feasible to use commercially.

Classification of electrically conducting materials

- Low resistivity electrically conducting materials possess few following additional properties:

i. Low-temperature coefficient

- Low-temperature coefficient means that the change in resistance with a change in temp. should be low.
- It is necessary to avoid variations in voltage drop and power loss with changes in temperature.

Classification of electrically conducting materials

ii. Ductility

- It is a property of the material that allows a material to be drawn or plastically deformed without fracture into thin wire.
- This property is important because conducting materials are required in different sizes and shapes
- so they should be ductile enough to be drawn into different shapes and sizes

Classification of electrically conducting materials

iii. Mechanical strength

- Mechanical strength of a material is required because in many applications conductors experience many stresses which may create some difficulties or maybe the complete failure of the devices
- There are the following stresses experienced by the materials:

1. Overhead line conductors used for transmission and distribution of electrical power are subject to stresses due to wind and their weight.

2. When the coils for winding of transformers, motors, and generators are made on formers, the conducting material is subject to mechanical stresses.

Classification of electrically conducting materials

iv. Resistance to corrosion

- In many applications conducting materials are used in outdoors.
- So, the materials should be such that it is not corroded when used in the outdoor atmosphere

Classification of electrically conducting materials

2. High resistivity conducting materials

- High resistivity materials or in other terms low conductivity materials are those materials that possess high resistivity or low conductivity properties.
- Insulators fall under this category.
- Carbon, Tungsten, nichrome, and magnain are high resistivity materials.
- High resistivity materials are used in all instruments where a high resistance value is required.
- High resistivity materials are used for making filaments for incandescent bulbs, resistance for heating devices, starters for electric motors, etc.

Classification of electrically conducting materials

- High resistivity electrically conducting materials possess few following additional properties:

i. Low-temperature coefficient

- For precision applications like making wire-wound precision resistance and resistance boxes and few others,
- it is a necessary requirement that the materials used should have very low-temperature coefficient resistance.
- If the temp. the coefficient is high then it may affect the accuracy of measurements of devices.

Classification of electrically conducting materials

ii. Ductility

- The material should be ductile because high resistance materials are required to be made for different shapes and sizes.
- For eg. In heaters, the starters material used should be thick in size and for precision wire wound resistors it should be thin.
- Therefore, the material should be capable of being drawn into a wire of different sizes and capable of being coiled.

Classification of electrically conducting materials

iii. High mechanical strength

- High resistivity materials are used as thin wires in some applications.
- The wire must be required to have high tensile strength so the wire doesn't break during operations.

iv. Do not oxide

- High resistivity materials are used in many heating appliances, so they should be able to withstand high temp.
- for a long time without oxidation.
- This is because if an oxide layer is formed on the material the amount of heat radiation will reduce.

Classification of electrically conducting materials

v. High melting point

- The materials should have a high melting point because in applications like starters for electrical motors the materials should withstand high temp for a long time without melting

Fermi-Dirac distribution function

- The Fermi energy is a concept in quantum mechanics usually refers to the energy difference between the highest and lowest occupied single-particle states in a quantum system of non-interacting fermions at absolute zero temperature.
- The highest energy level that an electron can occupy at the absolute zero temperature is known as the **Fermi Level**.
- The Fermi level lies between the valence band and conduction band because at absolute zero temperature the electrons are all in the lowest energy state.
- The value of the Fermi level at absolute zero temperature (−273.15 °C) is known as the **Fermi energy**.
- It is also the maximum kinetic energy an electron can attain at 0K.
- Fermi energy is constant for each solid.

Fermi-Dirac distribution function

What is Pauli Exclusion Principle?

- The Pauli exclusion principle states that in a single atom no two electrons will have an identical set or the same quantum numbers (n, l, m_l, and m_s).
- To put it in simple terms, every electron should have or be in its own unique state (singlet state).
- There are two salient rules that the Pauli Exclusion Principle follows
  1. Only two electrons can occupy the same orbital.
  2. The two electrons that are present in the same orbital must have opposite spins or they should be antiparallel.

Fermi-Dirac distribution function

- FDD function applies to fermions, particle with half integer spin which must obey the **Pauli Exclusion Principle.**
- Each type of distribution function has a normalization term multiplying the exponential in the denominator which may be temperature dependent.

$$f(E)=\frac{1}{e^{\left(\frac{E-E_f}{KT}\right)}+1}$$

- The meaning Fermi energy is mostly clearly seems setting T = 0.
- At absolute zero probability is 1 for energy less than Fermi energy.

$$E_f = \frac{h^2}{8m}\left(\frac{3N}{\pi V}\right)^{2/3}$$

$\frac{N}{V}$=n is number of Fermi electron in your system per unit volume

Fermi-Dirac distribution function

Examples

1. At what temperature we can expect a 10% probability the electrons in silver have an energy which is 1% above the Fermi energy. Given Fermi energy is 5.5eV.
2. Calculate the Fermi velocity of electrons in potassium if its Fermi energy is 2.1eV. Calculate number of electrons per unit volume.
3. Silver has exactly 1 conduction electron per atom. Use the density of silver $1.05 \times 10^4 kg/m^3$ and the molar mass of 107.87 g/mol to find the density of conduction electrons in silver. Use this to calculate the Fermi energy of silver and Fermi speed of silver.

Exercises

1. The Fermi energy of lithium is 4.72eV at T = 0K. Calculate the number of electrons per unit volume.
2. Calculate the temperature at which there is 1% probability that a state with an energy of 2eV is occupied. Given Fermi energy 1.5eV.